

# Geometry-Aware JEPA for EEG

A controlled frozen-transfer study on TUAB

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Hack the World(s) – EEG Track   ·   Team **HelloWorlds**

## Contribution

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# Our contribution, in one slide

1. **A strong frozen baseline.** SIGReg-ambient JEPA matches the LeJEPA/Laya recipe's value on our (pipeline-helped) setup – not a controlled head-to-head.
2. **Nothing beats it.** Ambient/tangent SIGReg & PEIRA, Graph-JEPA, Fourier-JEPA – none beats ambient SIGReg in balanced accuracy (gaps  $\leq 0.013$ ).
3. **Latent geometry is the contribution.** Ours separates normal/abnormal  $\sim 2\times$  better than the random-init encoder\*, visualised with **de Surrel's AIRM Riemannian t-SNE** (the right metric for SPD/EEG covariances). Tangent sharpens the *geometry* (best cov-latent silhouette), not the task.
4. **Rigor over spin.** The limits we report – *decodable*  $\neq$  *clustered*, sustained vs transient, AIRM edge within noise. An honest negative beats a fragile positive.

\* *Random-init encoder* = identical architecture, weights never trained, encoder frozen; only the linear probe is fit – it isolates the SSL contribution.

# Motivation

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## The question we actually asked

- **EEG foundation models** are everywhere – but most report *fine-tuned* numbers. What survives when you **freeze** the encoder?
- **Self-supervised JEPA** avoids collapse with an *anti-collapse regulariser*. The EEG covariance lives on a curved **SPD manifold**, not in flat Euclidean space.
- **Our pre-registered hypothesis**: making anti-collapse *geometry-aware* (act in the SPD **tangent**) and *distribution-free* (**PEIRA**) beats a plain ambient baseline.
- **What we ship**: a controlled study with an *honest* answer – a strong frozen probe, a clean negative result with a geometric mechanism, and label/data-efficiency curves.

## Data

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## Data: TUH Abnormal EEG (TUAB)

- **Task:** binary *normal* vs *abnormal* EEG, recording level.
- **Signal:** 19 channels @ 200 Hz; we sample 10 s windows (2,000 samples), per-channel z-scored.
- **Standard split:** 2,717 train / 276 eval recordings, **patient-disjoint** (2,076 vs 253 patients, overlap = 0 – verified).
- **Pretraining:** SSL reads only *unlabeled* train recordings; labels enter at the probe alone.

### Why we don't chase accuracy

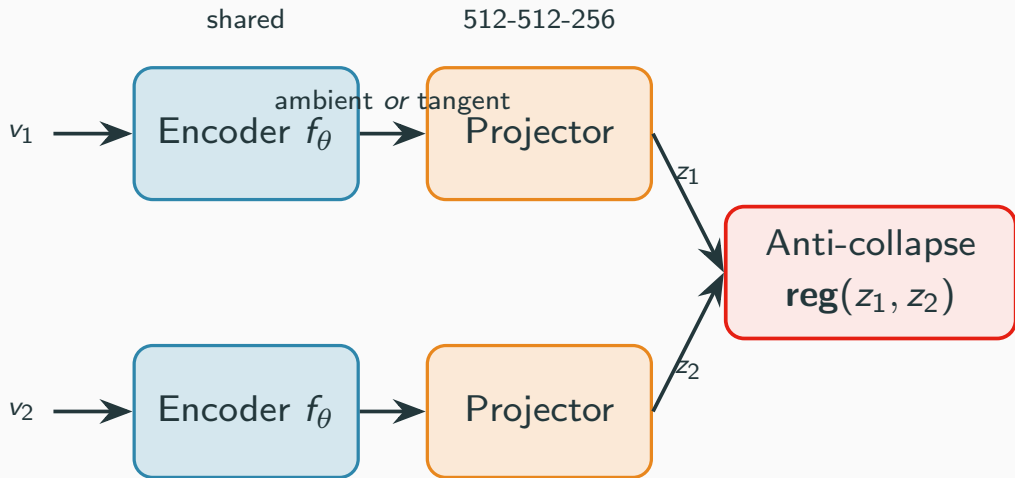
TUAB labels come from clinical reports and are known to be noisy; we are not aware of a published label-noise / Bayes-ceiling estimate. **Sub-1-point gaps without multi-seed CIs are not meaningful** – so we report *balanced accuracy on the full split*, 3 seeds, with error bars.

# Architecture

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## Two view and 1D convolutional encoder



## What we tried

| regulariser \ space                | ambient (Euclidean)          | tangent (SPD)           |
|------------------------------------|------------------------------|-------------------------|
| <b>VICReg</b> (variance/cov)       | C0 reference                 | —                       |
| <b>SIGReg</b> (Gaussianity slices) | C1 <i>Laya-like baseline</i> | C2 geometry-aware       |
| <b>PEIRA</b> (distribution-free)   | C3                           | C4 <i>ex-hypothesis</i> |

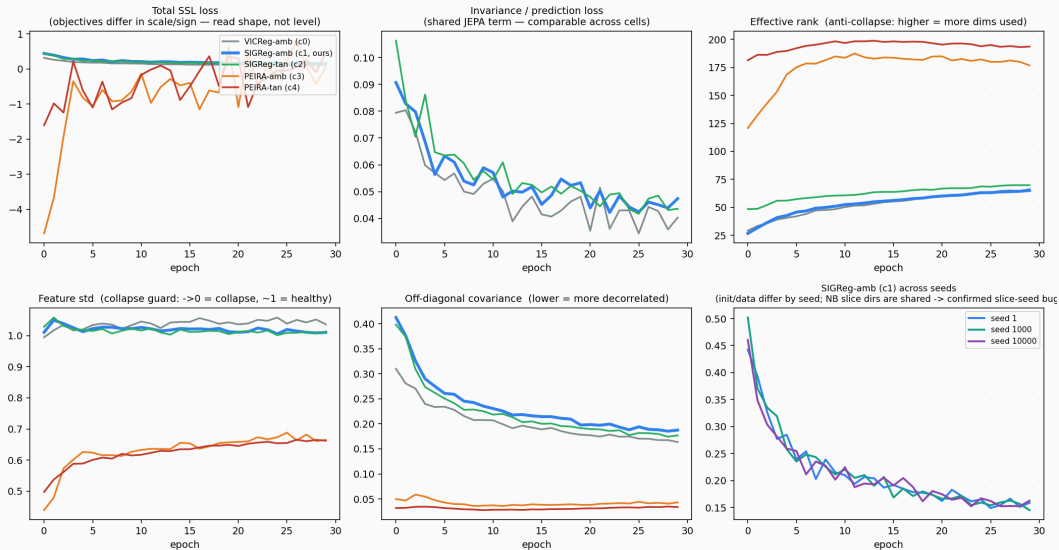
- **SIGReg** (LeJEPa/BCS): tests random 1-D slices for Gaussianity – assumes an *isotropic* target.
- **PEIRA**: distribution-free anti-collapse – no target distribution to mis-specify.
- **Hypothesis**: the geometry-aware / distribution-free cells (C2, C4) beat the ambient baseline C1.
- **GraphJEPa**
- **FourierJEPa**

## Results

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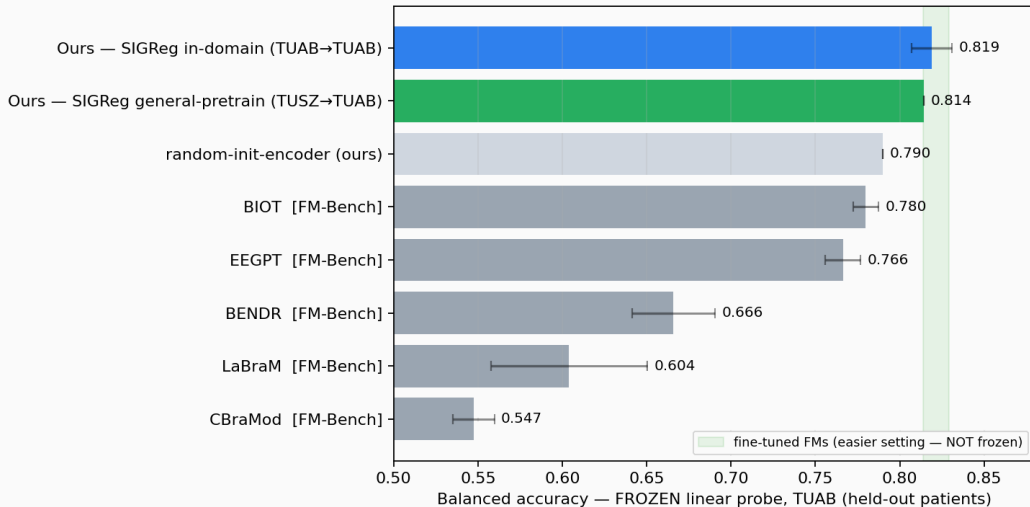
# SSL training curves (all cells healthy)

EEG-JEPA SSL training curves — 5 regularizer cells, 30 epochs, TUAB pretrain (seed 1)



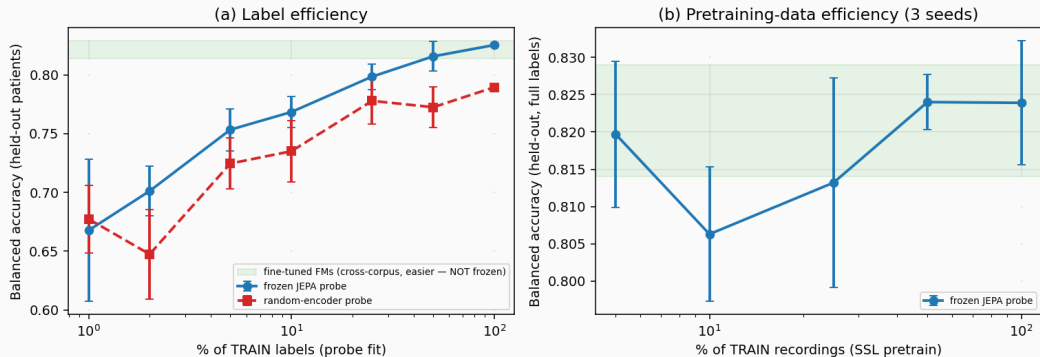
# Frozen & in-domain: we hold where frozen FMs fall

Frozen head-to-head on TUAB — FM rows: EEG-FM-Bench (arXiv 2508.17742, )



# The value of self-supervision (label & data efficiency)

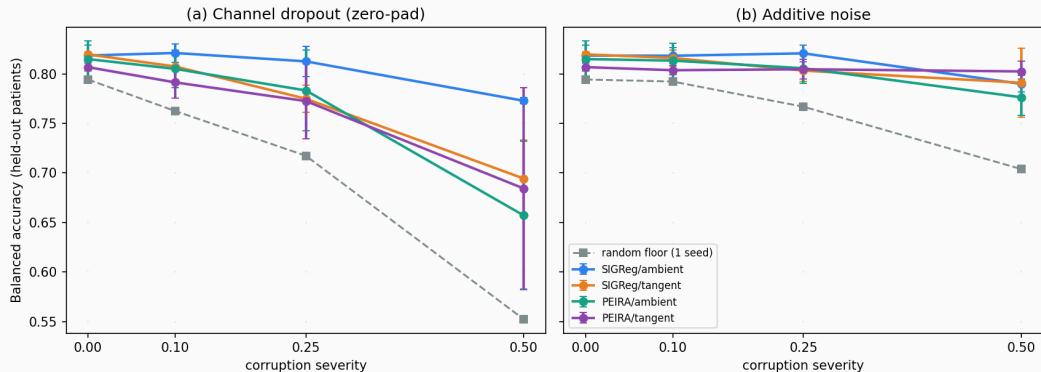
Value of self-supervision — frozen in-domain EEG-JEPA on TUAB



(a) The money plot: *JEPA @ 25% labels  $\approx$  random-init encoder @ 100%* —  $\sim 4\times$  label efficiency,  $\sim +0.04$  BalAcc. (b) Probe is stable down to small pretraining-data fractions.

# Beyond accuracy: frozen-probe robustness (3-seed)

Frozen-probe robustness on TUAB (3-seed mean  $\pm$  std) — ambient SIGReg is the most dropout-robust



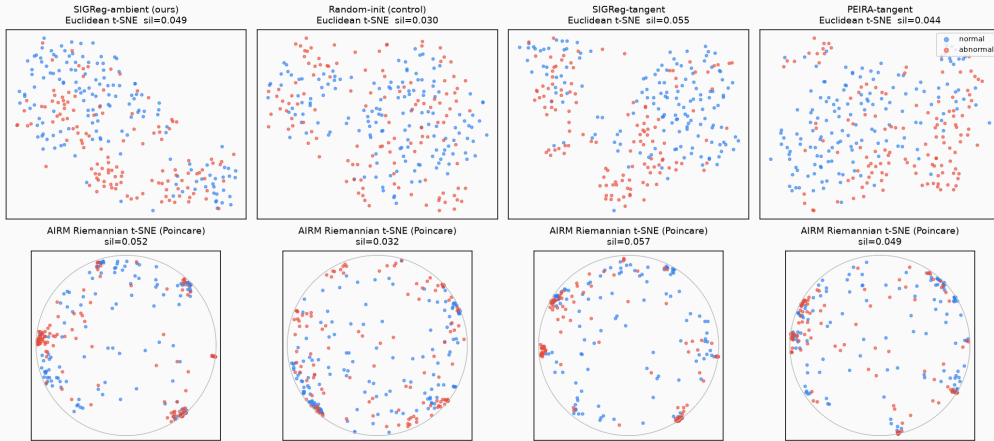
# Visualization

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# Manifold-aware latent geometry (AIRM vs Euclidean)

Frozen SPD latents on TUAB eval — Euclidean (tangent) t-SNE vs AIRM Riemannian t-SNE  
silhouette is full-dim (Euclidean tangent vs AIRM geodesic); 2-D maps are illustrative



<sup>†</sup> The tangent payoff is *SIGReg-specific*: SIGReg-tangent gives the best cov-latent silhouette on both TUAB (0.057) and TUSZ (0.085);

# Manifold viz across TUH tasks – sustained vs transient

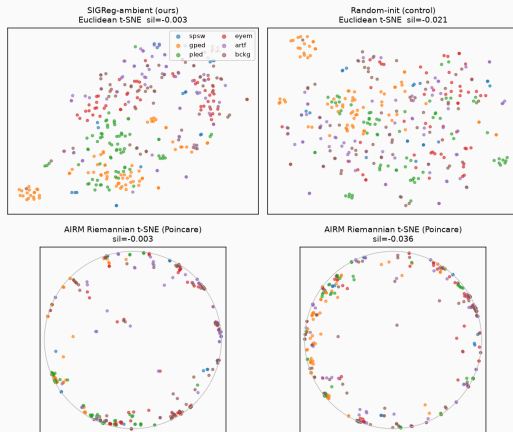
## TUSZ – 5 classes; SIGReg-tangent separates best

Frozen SPD latents on TUSZ eval – 5 seizure types (cross-task, same-site)  
Euclidean (tangent) t-SNE vs AIRM Riemannian t-SNE; silhouette is full-dim



## TUEV – 6 event types (decodable, not clustered)

Frozen SPD latents on TUEV eval – 6 event classes (cross-task, same-site)  
Euclidean (tangent) t-SNE vs AIRM Riemannian t-SNE; silhouette is full-dim



DECODABLE (frozen probe BA ~0.40, >> chance 0.17; ours > random) but NOT CLUSTERED (silhouette ~0)  
transient sub-second events diluted by the 10 s window -- silhouette measures clustering, not decodability

task (#cls)

$sil_{\text{Eucl}}$

$sil_{\text{AIRM}}$

• Clusters **sustained** phenomena (abnormality, seizures), not

## Caveats we disclose to the jury

- **Report 0.819, not 0.833.** The single-seed 0.833 (SIGReg-ambient, seed 1) was the *top* of that cell's range; the 3-seed mean is 0.819.
- **SIGReg slice-seed bug (found, fixed).** BCS seeded slice directions per-step, so the 3 SIGReg seeds shared one slice schedule – error bars *understate* variance. Now fixed (run-seeded RNG); the reported numbers predate the fix, not re-run.
- **PEIRA tangent (0.807) is not evidence against wrapped-Gaussian.** PEIRA is distribution-free  $\Rightarrow$  immune to the isotropic-target mis-specification  $\Rightarrow$  it never tests that hypothesis. 0.807 vs 0.815 is within noise.
- **Bootstrap CI (illustrative, not yet run).** A bootstrap over the 276 eval recordings would plausibly give a  $\pm 0.02$ – $0.03$  CI – wider than the max inter-cell gap (0.013) – supporting “benchmark-resolution-limited”. The resampling is not yet committed.
- **Preprocessing differs** from the published FM rows – the head-to-head is indicative, not a controlled re-run of their pipelines.

## Limitations & Future Work

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## Limitations & Future Work

- **Single dataset.** Specialization  $\neq$  generality. The honest generality test is frozen cross-dataset transfer (TUEV/Sleep-EDF) – started, but left as future work rather than over-claimed.
- **Underpowered null.**  $n=3$  seeds; the inter-cell gaps ( $\leq 0.013$ ) are below a plausible eval bootstrap CI ( $\pm 0.02$ – $0.03$ , not yet computed). The honest statement is “*benchmark-resolution-limited*”, not “geometry is dead”.
- **The principled fix.** A wrapped-Gaussian SIGReg on the *channel* covariance, AIRM tangent at the running Fréchet mean –  $\sim 40$  LOC + one re-pretrain, with real eigh-backward numerical risk.
- **Where geometry should pay off.** Robustness, calibration, and transfer under montage/threshold shift – *not* a balanced-accuracy race.

## Conclusion

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# Conclusion

- A **controlled frozen-transfer study**, not a leaderboard entry.
- A **strong frozen probe** (0.819 BA,  $\sim 0.89$  AUROC) that holds where published FMs collapse when frozen – with the random-init-encoder floor honestly disclosed.
- A **clean, pre-registered negative result**: *where* anti-collapse acts (ambient vs SPD-tangent) and *which* regulariser do not change frozen-probe accuracy on TUAB – with a geometric reason why.
- **The null is multi-axis**: geometry/PEIRA help neither accuracy, nor calibration, nor robustness (3 seeds) – and it holds even *general*-pretrained (TUSZ  $\rightarrow$  frozen TUAB, 0.814).
- **Honesty as the headline**: an honest negative beats a fragile positive.

Built on a trimmed eb\_jepa vendor. Team **Hello Worlds**.

## References

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[github.com/facebookresearch/eb\\_jepa](https://github.com/facebookresearch/eb_jepa). (*our trimmed code vendor*)